Curriculum Vitae - Li Hongmin

Email: lihongmin[at]edu.k.u-tokyo.ac.jp

Primary Affiliation: Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo (Laboratory)

Guest Affiliation: Frith Lab, The University of Tokyo (Laboratory)

Research Interests

- AI-Automated Scientific Workflows
- AI for Science and Scientific Discovery
- AI Research and Evaluation
- Bioinformatics and Biological Sequence Analysis
- Biomolecular Sequence Design and Optimization

Education

- **2019.04 —2022.03** | **Ph.D. in Computer Science (Information Systems Engineering)** —University of Tsukuba
- **2017.04 —2019.03** | **Master's in Computer Science (Information Systems Engineering)** —University of Tsukuba
- **2011.09 —2015.07** | **Bachelor's in Electronic Information Engineering** —Ningxia University

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Academic & industry appointments

- **2026.4 —Present** | **Researcher** —Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo; Laboratory
- **2026.4 —Present** | **Guest Researcher** —Frith Lab, The University of Tokyo; Laboratory
- **2026.6 —Present** | **Founder** —AlphaScience Lab (Research community); Website
- **2023.5 —2026.3** | **Postdoctoral Researcher** —The University of Tokyo
- **2022.10 —2023.5** | **Machine Learning Engineer** —HAOMO.AI —Autonomous-driving project · 蓝色空间领航者 (A super high salary project)（类似华为天才少年计划）
- **2022.4 —2022.10** | **Postdoctoral Researcher** —The University of Tokyo; Details
- **2021.7 —2021.9** | **Machine Learning Engineer (Internship)** —HAOMO.AI

Research funding & computing support

- **2026.06** | **Google Cloud TPU Builders Award** —Cloud computing support · USD 7,500 in Google Cloud computing credits
- **2026.04** | **Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of an Input Data Differentiable Integrated Framework (ID3) for Biomolecular Sequence Design · ¥4,550,000; Project info
- **2025.10** | **Google research support** —Biological sequence optimization · USD 30,000
- **2024.4** | **Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of a Large-Scale Language Model Integrating RNA Sequences and Text · ¥4,420,000; Project info

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Publications

Books

- **2026.08** | Hongmin Li. "*Academic Collapse Does Not Happen All at Once: A Conditional Theory of Epistemic Cascades, Expertise Cliffs, and Institutional Recoupling*" . Book · Independently published · 310 pp. · ISBN 979-8192323182. Amazon

Preprints

- **2026.09** | Hongmin Li, Wanli Zhao. "*Where Scientific Search Agents Fail: Decision-Checkpoint Auditing of Exposure and Inspection Attempts*" . arXiv preprint · 2609.38670. arXiv
- **2026.09** | Hongmin Li. "*The Hodge Conjecture for Odd-Degree Fermat Fourfolds*" . ResearchGate preprint. ResearchGate
- **2026.09** | Hongmin Li. "*Low-t Coefficients of the Valley Delta Conjecture*" . ResearchGate preprint. ResearchGate
- **2026.09** | Linghu Fan, Hongmin Li. "*Resolutions of Linear p-Cyclic Quotient Singularities*" . arXiv preprint · 2609.07182 [math.AG]. arXiv
- **2026.06** | Hongmin Li. "*The Calibration Turn in AI-Assisted Research: A Conceptual and Methodological Framework for Evidence-Licensed Claims*" . arXiv preprint · 2606.31273 [cs.LG]. arXiv, Code & artifacts
- **2026.05** | Hongmin Li. "*Targeted Tests for LLM Reasoning: An Audit-Constrained Protocol*" . arXiv preprint · 2605.11599 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*Separating Shortcut Transition from Cross-Family OOD Failure in a Minimal Model*" . arXiv preprint · 2605.12945 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*A Controlled Counterexample to Strong Proxy-Based Explanations of OOD Performance: in a Fixed Pretraining-and-Probing Setup*" . arXiv preprint · 2605.11554 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*FastUMAP: Scalable Dimensionality Reduction via Bipartite Landmark Sampling*" . arXiv preprint · 2605.11428 [cs.LG]. arXiv
- **2025.12** | Bian Bian, Yiming Zhang, Hongmin Li, Jiuzhou Zhong, Yutaka Saito. "*Large generative mRNA language foundation model for efficient coding sequence generation and design with mRNA-GPT*" . bioRxiv. Preprint
- **2023.9** | Hao Wu, Xiangjun Pei, Takashi Oguchi, Hongmin Li, Zifei Wang, Zhihao He, Runqiu Huang. "*An Integrated Physical Approach to Earthquake-Induced Landslide Susceptibility Incorporating Geological Structure: A Case Study of the Diexi Catchment, Sichuan, China*" . SSRN preprint. SSRN

Journal Articles

- **2026.09** | Hongmin Li, Goro Terai, Takumi Otagaki, Kiyoshi Asai. "*Gradient-based Optimization for mRNA Sequence Design*" . Bioinformatics. Journal / DOI, Preprint, Code
- **2023.1** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*LSEC: Large-scale spectral ensemble clustering*" . Intelligent Data Analysis. arXiv, Code
- **2022.8** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Divide-and-conquer based Large-Scale Spectral Clustering*" . Neurocomputing. Preprint, Code
- **2022.7** | Meng Huang, Xiucai Ye, Hongmin Li, Tetsuya Sakurai. "*Missing Value Imputation With Low-Rank Matrix Completion in Single-Cell RNA-Seq Data by Considering Cell Heterogeneity*" . Frontiers in Genetics. DOI
- **2020.7** | Xiucai Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. "*An Oversampling Framework for Imbalanced Classification Based on Laplacian Eigenmaps*" . Neurocomputing. Link
- **2019** | Xiucai Ye, Hongmin Li, Tetsuya Sakurai, Pei-Wei Shueng. "*Ensemble Feature Learning to Identify Risk Factors for Predicting Secondary Cancer*" . International Journal of Medical Sciences. DOI

Conference Papers

- **2020.11** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Ensemble Learning for Spectral Clustering*" . ICDM 2020. PDF, Code
- **2020.7** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Hubness-based Sampling Method for Nyström Spectral Clustering*" . IJCNN 2020. Link
- **2019.8** | Xiucai Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. "*Distributed Collaborative Feature Selection Based on Intermediate Representation*" . IJCAI 2019. Link
- **2018.10** | Xiucai Ye, Hongmin Li, Tetsuya Sakurai, Zhi Liu. "*Large Scale Spectral Clustering Using Sparse Representation Based on Hubness*" . CBDCom 2018. Link, Code

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AI-generated mathematics (rStation)

Mathematical manuscripts generated with AI assistance and publicly posted on rStation.

- **2026.10.27** | "*Exponentially Many Barnette Graphs with the Global H^{+-} Property*" . rStation r/math, math_1 v2. https://rstation.ai/problems/math/papers/1/v2
- **2026.10.26** | "*A Positive Reciprocal-Product Criterion for the Riemann Hypothesis*" . rStation r/riemann_hypothesis, riemann_hypothesis_42 v2. https://rstation.ai/problems/riemann_hypothesis/papers/42/v2
- **2026.10.26** | "*Generalized Thomas Sections and Density-Corrected Residue Selectors*" . rStation r/hodge_conjecture, hodge_conjecture_39 v2. https://rstation.ai/problems/hodge_conjecture/papers/39/v2
- **2026.10.26** | "*A Single Modular Character Is Complete for SYM∘LTF Counting*". rStation r/computer_science, computer_science_1 v2. https://rstation.ai/problems/computer_science/papers/1/v2
- **2026.10.25** | "*Truth-Table Quotients Do Not Close One-Tape MCSP Magnification: Small Fibers, Exact Cosets, and a Defect-Inflation Dichotomy*" . rStation r/p_vs_np, p_vs_np_1 v1. https://rstation.ai/problems/p_vs_np/papers/1/v1
- **2026.10.25** | "*Tight Ordered Width for Exponential-Threshold MCSP*" . rStation r/computer_science, computer_science_2 v1. https://rstation.ai/problems/computer_science/papers/2/v1
- **2026.10.25** | "*Concentration of Random Tensor Sumsets*" . rStation r/math, math_2 v1. https://rstation.ai/problems/math/papers/2/v1
- **2026.10.23** | "*Cutoff-free positivity of localized Weil forms through the (log 11)/2 threshold*" . rStation r/riemann_hypothesis, riemann_hypothesis_45 v1. https://rstation.ai/problems/riemann_hypothesis/papers/45/v1
- **2026.10.22** | "*Balanced Cyclic Prym Families and the Generalized Hodge Conjecture for All Powers*". rStation r/hodge_conjecture, hodge_conjecture_38 v1. https://rstation.ai/problems/hodge_conjecture/papers/38/v1
- **2026.10.22** | "*Moving Apolar Constraints on Hodge Cycles of the Fermat Sextic Fourfold*" . rStation r/hodge_conjecture, hodge_conjecture_41 v1. https://rstation.ai/problems/hodge_conjecture/papers/41/v1
- **2026.10.19** | "*A small-diagonal transfer from a degree-33 Fermat character to the quartic-CM determinant of a Prym eightfold, and the Hodge conjecture for its powers under explicit quotient hypotheses*" . rStation r/hodge_conjecture, hodge_conjecture_20 v3. Conditional on explicit quotient hypotheses stated in the paper. https://rstation.ai/problems/hodge_conjecture/papers/20/v3

Conference presentations

- **2026.06.09** | **Poster presentation —JST CREST BioDX Area Meeting** —5th area meeting · Numazu, Japan
- **2025.11** | **Poster presentation —JST CREST BioDX Interim Symposium** —Event
- **2025.8** | **Oral presentation —Input Data Differentiable Designer (ID3) at RNA Informatics Dojo 2025** — Early presentation of the method later developed into the preprint "Gradient-based Optimization for mRNA Sequence Design" ; Event, Paper, Code
- **2024.10** | **Poster presentation —FastUMAP at the Asia-Pacific Bioinformatics Joint Conference 2024** —Early conference presentation of the work later developed into the FastUMAP preprint; Event, Paper
- **2024.8** | **Oral presentation —RNA Informatics Dojo 2024**
- **2024.8** | **Participation —IJCAI 24 (Jeju)**
- **2024.6** | **Participation —WCCI 24 (Yokohama)**
- **2024.6** | **Participation —25th Annual Meeting of the RNA Society of Japan**
- **2023.8** | **Oral presentation —RNA Informatics Dojo 2023**
- **2022.8** | **Oral presentation —RNA Informatics Dojo 2022**
- **2022.3** | **Presentation —JST SPRING recipients event**
- **2021.12** | **Oral presentation —CollaboTICS 2021**
- **2019.6** | **Poster presentation —G20 Digital Science Now Symposium**

Peer Review Activities

- **2026.07** | **Conference on Neural Information Processing Systems (NeurIPS)**
- **2026.07** | **Neurocomputing**
- **2026.05** | **Journal of King Saud University - Computer and Information Sciences**
- **2026.01** | **The Journal of Supercomputing**
- **2025** | **International Journal of Machine Learning and Cybernetics**
- **2024.11** | **Pattern Analysis and Applications**
- **2024.11** | **Knowledge and Information Systems**
- **2024.10** | **Neurocomputing**
- **2024.09** | **The Visual Computer**
- **2024.03** | **International Journal of Machine Learning and Cybernetics**
- **2022.11** | **Transactions on Pattern Analysis and Machine Intelligence**
- **2022.10** | **Briefings in Functional Genomics**
- **2021–2022** | **Pattern Recognition**

Honors & fellowships

- **2021.10** | **Doctoral research support —JST SPRING** —Support for Pioneering Research Initiated by the Next Generation
- **2020.10** | **AETA Earthquake Prediction AI Algorithm Competition 2019 —Second Prize** —Second-place recognition for an AI model predicting earthquakes from real-time monitoring data · awarded in 2020; News, Slides
- **2019.10** | **3rd Analysys International Algorithm Competition —Special Award** —Special recognition in the website page-view and unique-visitor prediction task
- **2018.10** | **IEEE International Conference on Cloud and Big Data Computing 2018 —Best Paper Award** —Awarded for "Large Scale Spectral Clustering Using Sparse Representation Based on Hubness"

Skills

Programming Languages

- Python, R, MATLAB, C/C++, Java

Machine Learning & Deep Learning

- PyTorch, TensorFlow, scikit-learn
- Spectral Clustering, Large Language Models
- Bioinformatics Applications

Computational Biology

- RNA Sequence Analysis
- mRNA Design and Optimization
- Single-cell RNA-seq Data Analysis